

LAB 13: NETWORK PACKET CAPTURE USING WIRESHARK

Objective:

To use Wireshark to capture and analyze data packets

Software Used:

Wireshark

Theory:

Wireshark

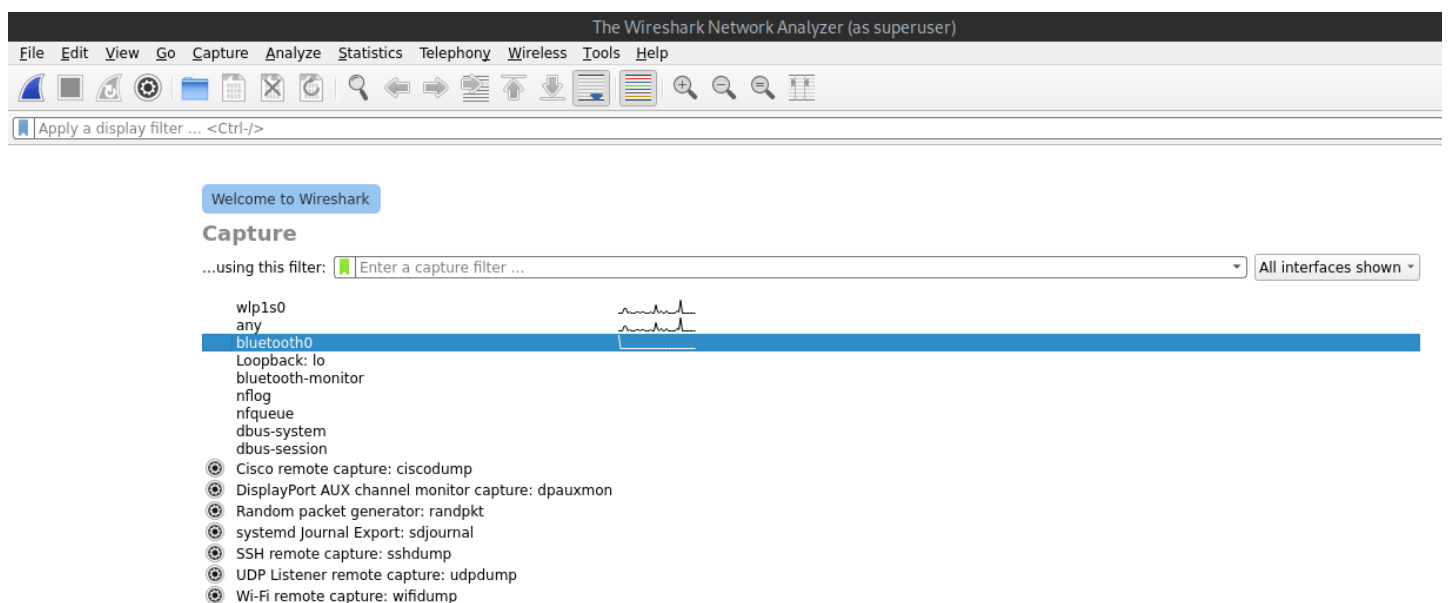
Wireshark is a free network protocol analyzer that runs on Windows, Linux/Unix, and Mac computers. It is a free open-source packet sniffer tool. It is used for network troubleshooting and communication protocol analysis. Wireshark captures network packets in real time and displays them in human readable format. It provides many advanced features including live capture and offline analysis, three-pane packet browser, coloring rules for analysis.

Packet Sniffer

Packet sniffer is a basic tool for observing network packet exchanges in a computer. As the name suggests, a packet sniffer captures (sniffs) packets being sent and received from/by our computer; it also stores and displays the contents of the various protocol fields in these captured packets. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on our computer, but never sends packets itself.

Starting Wireshark

When we run the Wireshark program, the Wireshark graphic user interface will be shown as the following Figure. Currently, the program is not capturing the packets.

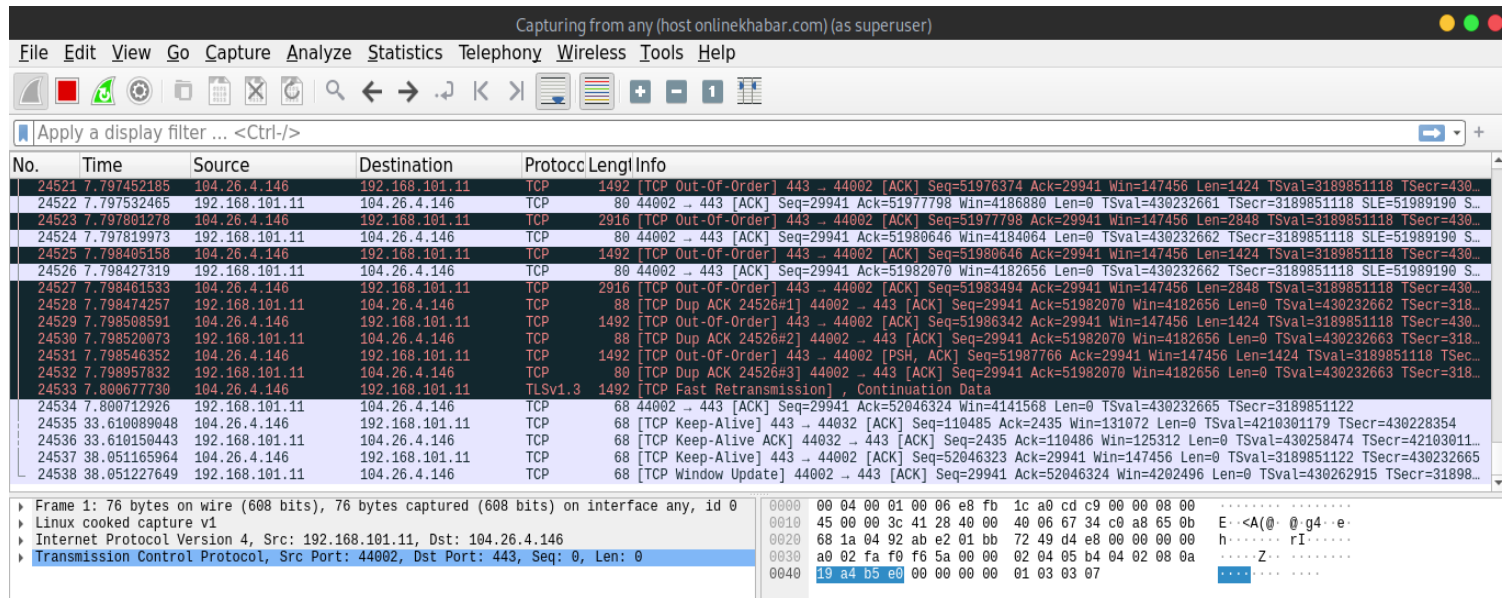


Now, we need to choose an interface. There are many interfaces available. I choose Wi-Fi remote capture interface. Once an interface is chosen, click start capturing packets i.e. leftmost icon from top left corner. Then, we will see the capturing of various network packets.

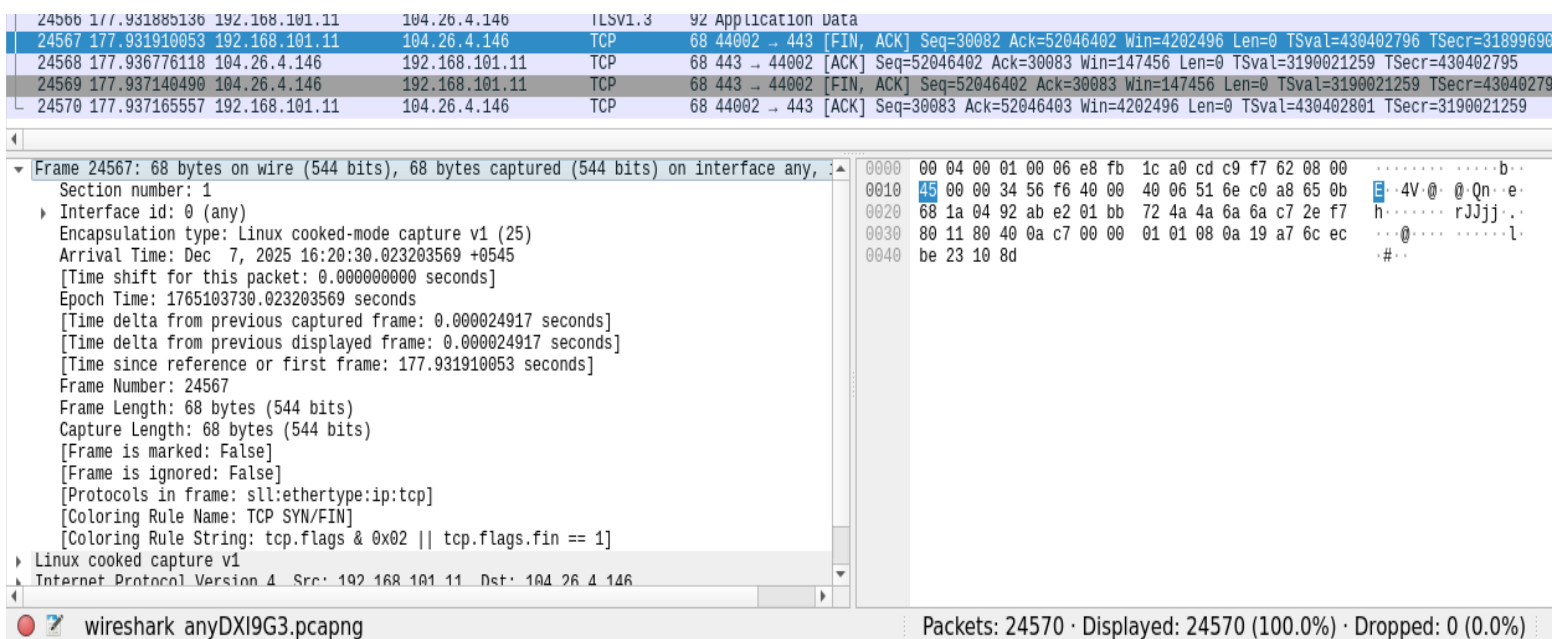
Now, do the followings to capture a network packet exchanged with a website:

1. Start up a browser to request to access any webserver (website).

2. Simultaneously, start the packets capture in Wireshark.
3. While the Wireshark is capturing packets, enter a URL (here, <https://onlinekhabar.com>) on the web browser and have that page displayed in the browser. The browser will contact the HTTP server at onlinekhabar.com and exchange HTTP messages with the server. The Ethernet frames containing these HTTP messages will be captured by Wireshark.



4. After the browser has displayed the onlinekhabar page, stop Wireshark packet capture. Now we have live packet data that contains all protocol messages exchanged. The HTTP message exchanges with the onlinekhabar.com web server should appear in the listing of packets captured. But there will be many other types of packets displayed as well.
5. Look for the IP address of the web server (onlinekhabar.com) in the source. Then we can see the information about that packet.



One HTTP/HTTPS packet clearly showing onlinekhabar.com

→ In the filter bar (top), type:

`tls.handshake.type == 1` or `http`

click the first TLS Client Hello or any HTTP packet

→ Expand → TLS

The image shows a Wireshark network traffic capture. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons. The filter bar at the top displays the filter `tls.handshake.type == 1` or `http`. The packet list pane shows several packets, with the selected packet (No. 3365) being a TLSv1.3 Client Hello. The packet details pane on the right shows the structure of the TLSv1.3 Record Layer: Handshake Protocol: Client Hello. The packet bytes pane at the bottom shows the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
4	0.063485326	192.168.101.11	104.26.4.146	TLSv1.3	674	Client Hello
76	0.445829319	192.168.101.11	104.26.4.146	TLSv1.3	519	Client Hello
89	0.450033196	192.168.101.11	104.26.4.146	TLSv1.3	519	Client Hello
2096	1.762906724	192.168.101.11	104.26.5.146	TLSv1.3	982	Client Hello
2100	1.766033224	192.168.101.11	104.26.5.146	TLSv1.3	982	Client Hello
2202	2.054049499	192.168.101.11	172.67.71.14	TLSv1.3	519	Client Hello
3365	2.647939494	192.168.101.11	104.26.4.146	TLSv1.3	674	Client Hello

Frame 3365: 674 bytes on wire (5392 bits), 674 bytes captured (5392 bits) on interface any

- Linux cooked capture v1
- Internet Protocol Version 4, Src: 192.168.101.11, Dst: 104.26.4.146
- Transmission Control Protocol, Src Port: 44032, Dst Port: 443, Seq: 1, Ack: 1, Len: 606
- Transport Layer Security
 - TLSv1.3 Record Layer: Handshake Protocol: Client Hello
 - Content Type: Handshake (22)
 - Version: TLS 1.0 (0x0301)
 - Length: 601
 - Handshake Protocol: Client Hello
 - Handshake Type: Client Hello (1)
 - Length: 597
 - Version: TLS 1.2 (0x0303)
 - Random: 1c9e3a259de2a81323e97192849c3da8e880b8b2ef3a8a6e0fc9cddb018345854
 - Session ID Length: 32
 - Session ID: 76144e7310f29aadd0b92abc40dc19c1a929e19382bc46b4c38c59092155ad01
 - Cipher Suites Length: 34
 - Cipher Suites (17 suites)
 - Compression Methods Length: 1
 - Compression Methods (1 method)

Conclusion

We successfully captured network packets using Wireshark while visiting <https://onlinekhabar.com>. The capture showed TCP and TLS handshakes, including the Server Name Indication (SNI) containing "onlinekhabar.com". Even a simple webpage load generates many background packets. This lab proves Wireshark is an effective tool for observing and understanding real-time network communication.